Hugo Pereira

hugoacpereira3011@gmail.com
(+351) 96 33 62 767
Lisbon, Portugal

Education

Instituto Superior Técnico, University of Lisbon 2020 - 2022
Master of Science in Aerospace Engineering (Control & Systems branch) GPA: 19 / 20

Technical University of Denmark 2020 - 2021
Exchange Student GPA: 11 / 12

Instituto Superior Técnico, University of Lisbon 2017 - 2020
Bachelor of Science in Aerospace Engineering GPA: 17 / 20

Research Experience

Carnegie Mellon University, ECE department *Pittsburgh, USA*
Visiting Researcher (Full-time) *Sep. 2026 - Present*

- Researching GPU-parallel solvers for model predictive control.
- Extending the parallel scan method to constrained optimal control problems, including its integration with operator splitting methods such as the ADMM and the progressive decoupling algorithm (PDA).
- Working on native CUDA C++ implementations of these solvers, tailored to specific GPU architectures.
- Supervisors: Prof. Soummya Kar (CMU), Prof. João Xavier (ISR/IST) and Dr. Pedro Lourenço (GMV).

Institute for Systems and Robotics, IST-UL *Lisbon, Portugal*
Graduate Student Researcher (Full-time) *Mar. 2022 - June 2023*

- Did research on singularity avoidance for spacecraft attitude control with control moment gyros (CMGs).
- Derived a novel convex MPC-based solution for CMG allocation.
- Developed a medium-fidelity 6 DoF simulator for spacecraft dynamics in LEO.
- Worked towards my master's thesis titled *"A Convex Allocation Framework for Singularity Avoidance in Control Moment Gyro Clusters"*, which was awarded the grade of 20 out of 20.
- Supervisors: Prof. Pedro Batista (ISR/IST) and Dr. Pedro Lourenço (GMV).

Publications

[3] Á. Palomino, F. Arribas, G. Landa, **H. Pereira** et al., *"A rapid response mission to Apophis: challenges and solutions through autonomous guidance, navigation and control"*, in Proceedings of the ESA GNC & ICATT Conference 2026, Seville, Spain (Sep-Oct. 2026) **[url]**.

[2] J. F. Briz, E. de Brito, M. Rubaga, M. Ferreira, L. Ferreira, **H. Pereira** et al., *"Tractor-Beam: An Experimental Testbed for Control of In-Orbit Servicing Systems with Varying Flexible Dynamics and Inertia"*, in Proceedings of the CEAS EuroGNC 2026 Conference, Madrid, Spain (May 2026), doi: 10.82124/CEAS-GNC-2026-062.

[1] **H. Pereira**, P. Lourenço, P. Batista, *"From singularity analysis to singularity avoidance: novel metric and convex allocation for spacecraft attitude control with control moment gyros"*, Acta Astronautica 225 (2024) 41-54, doi: 10.1016/j.actaastro.2024.08.047.

[**MSc thesis**] *"A Convex Allocation Framework for Singularity Avoidance in Control Moment Gyro Clusters"*, Instituto Superior Técnico, 2022.

TALKS

- *"A rapid response mission to Apophis: challenges and solutions through autonomous guidance, navigation and control"*. 12th International Workshop and Advanced School on Spaceflight Dynamics and Control, UBI Covilhã, 18-20 May 2026.

WORK EXPERIENCE

GMV *Lisbon, Portugal*

Control Systems Engineer (Full-time) *July 2023 - Present*

- Responsible for the design and development of the attitude controllers for the Rapid Apophis Mission for Space Safety (RAMSES) through the PDR, CDR and TRR phases; Carried out control synthesis, using LQR and pole-placement techniques, and frequency-domain analyses for all the control modes of the GNC (S/C detumbling, Sun pointing operations during Safe mode, RW and RCT control for attitude stabilization and slew maneuvers, and wheel off-loading); Compared with those used on HERA, the attitude controllers I developed resulted in significant pointing improvements and reduced propellant consumption, enabling extended periods in proximity operations; Co-developed attitude navigation solutions (gyro-stellar & gyroscopic torque estimator), FDIR algorithms for thruster failure detection, and actuator management functions; Performed extensive Monte Carlo simulations across multiple representative in-orbit scenarios and conducted unit tests for several GNC functions.
- Co-developed the attitude controllers for the ARIEL telescope, using structured $\mathcal{H}_\infty$ control, through the PDR and CDR phases; Designed a solution for attitude estimation by combining star tracker data with fine guidance sensor measurements; Derived guidance profiles for downlink communications based on Chebyshev polynomials.
- Worked as a system engineer for the IOANT project during the development of a testbed for single-axis attitude control of a spacecraft equipped with flexible appendages; Co-developed an in-house reaction wheel, designed the avionics for the platform based on Raspberry Pi and Wi-Fi communication, handled most of the hardware procurement, and took responsibility for the development of a digital twin using MATLAB/SIMULINK; Carried out MIL, SIL, PIL, and HIL tests.
- Participated in several stages of research-oriented ESA-funded activities on spacecraft collision avoidance (literature review and requirements definition) and space launchers (derivation of flight dynamics models and synthesis of convex optimization methods for thruster allocation).
- Co-supervised and oversaw the work of five interns on the topics of convex optimization, control theory, and system identification applied to flexible space structures.

Department of Electrical and Computer Engineering, IST-UL *Lisbon, Portugal*

Invited Teaching Assistant (Part-time) *Sep. 2024 - Jan 2025*

- Invited lecturer for the Convex Optimization course, delivering 6 hrs/week of problem-solving sessions to MSc classes. Provided support during the project assignments and graded the final exams. Received a final rating of 8.94 / 9.00 from the students.

UAVision *Torres Vedras, Portugal*

Summer Intern (Full-time) *June 2020 - July 2020*

- Short-term internship where I contributed to the design of a signal demodulation program for an Emergency Position Indicating Radio Beacon. The system, developed in collaboration with other interns, was sketched in GNU Radio and written in Python.

EXTRACURRICULAR ACTIVITIES

Formula Student Lisboa *Lisbon, Portugal*

Autonomous Systems Engineer (Part-time) *Sep. 2021 - Sep. 2022*

- Developed autonomous driving algorithms, using ROS/C++, for a driverless Formula car. The algorithms used in the pipeline gathered various state-of-the-art concepts, including machine learning for point cloud processing, state estimation via Kalman filtering, simultaneous localization and mapping (SLAM), and model predictive control (MPC).
- Specifically, I implemented new data association methods for SLAM, developed an iterative closest point (ICP) method for point cloud alignment, worked on point cloud processing algorithms, and designed and fine-tuned longitudinal and lateral nonlinear controllers for path following.
- Achieved 3rd place at Formula Student Germany 2022 and 1st at Formula Student Spain 2022 (driverless categories).

Danish Student Association for Rocketry *Copenhagen, Denmark*

Avionics Systems Engineer (Part-time) *Nov. 2020 - Oct. 2021*

- Part of the team that established the former world altitude record for bi-liquid propulsion rockets built by college students. Apogee: 6545 m AGL.
- Led the development of the rocket's telemetry system. Established a communication link based on LoRa technology that operated at a 9 km range. Designed, tested, and debugged PCBs for radio communication, using KiCad, within an STM32-based flight computer architecture.
- Carried out robust flight simulations using the OpenRocket software, which resulted in a predicted apogee error of less than 3%.
- Achieved 1st place at the European Rocketry Challenge 2021 (bi-liquid propulsion category).

Aerospace Engineering Students Group | AeroTéc *Lisbon, Portugal*

Control Systems Engineer (Part-time) *Oct. 2018 - June 2020*

- Built a UAV from scratch over the course of two years. Initially, I was involved in the design, 3D modelling, and construction of the UAV. At a later stage, I took part in the development of the flight controller, which involved PID control, filtering, path planning, sensor fusion, and system identification. The simulations were conducted in MATLAB/SIMULINK and the flight code was implemented on an RPi.
- Gave workshops about rocketry to high school students. These consisted of a theoretical lesson followed by a hands-on experience that included planning, building, and launching a small rocket made from cheap conventional materials.

COMPLEMENTARY TRAINING

- *Inter-Agency GNC V&V Workshop (IAGNC 2025), ISAE-SUPAERO, July 2025*. Short training course on H_∞ control and dynamic systems modeling and verification applied to spacecraft attitude control.
- *Ladybird Guide to Spacecraft Communications Training Course, 2022, ESA Academy*. Selected to attend a remote spacecraft communications course featuring live sessions given by ESA experts.
- *Summer course in Robotics, University of Twente, Aug. 2019*. On-site summer camp in robotics at the University of Twente in the Netherlands. This consisted of a hands-on experience with real-time robotics systems and included a group project centered on Arduino manipulation.

SKILLS SUMMARY

Languages	English (C2, TOEFL: 118/120), Portuguese (Native)
Programming	C++, C, CUDA, MATLAB, Python
Platforms	Linux, Windows, ROS, STM32, PIC24, Arduino
Software	SIMULINK, KiCad, OpenRocket, SolidWorks
Tools	Git, LaTeX, Microsoft Office, Jira
Miscellaneous	Soldering

AWARDS

- Academic Excellence, Instituto Superior Técnico, 2021/2022
- Outstanding Performance at EuRoc 2021, DTU Blue Dot, Oct. 2021
- Academic Merit, Instituto Superior Técnico, 2017/2018, 2019/2020
- Best Student Award, Escola Secundária Marquesa de Alorna, 2015/2016, 2016/2017

FELLOWSHIPS

- Selected for the CMU Portugal Visiting Students Program (2026 call).
- Awarded an FCT research scholarship under the EMSO-PT project (Call BL264, 2023).

VOLUNTEERING

Formula Student Portugal, FSPT *July 2025 & July 2026*

- Acted as a design judge for the driverless cars during the 2025 and 2026 editions. I was responsible for evaluating autonomous navigation solutions, including SLAM implementation and Kalman filter design.

Erasmus Student Network, DTU section *Feb. 2021 - June 2021*

- Member of an international team of volunteers whose goal is to facilitate the integration of international students in the local community.
- Organized social events and provided general support and guidance for exchange students.